Healthcare Risk Prediction — Machine Learning vs Neural Networks

Problem Understanding

- **Problem Description:**
  The objective is to predict diabetes risk in patients using their health data, such as age, BMI, blood pressure, exercise habits, and smoking status. The challenge is to develop models that can accurately classify patients as diabetic or non-diabetic and to compare the performance of a traditional Machine Learning model with a Neural Network.
- **Importance of Prediction:**
  Early prediction of diabetes is crucial for preventing serious health complications such as heart disease, kidney failure, and vision problems. By identifying high-risk patients in advance, healthcare providers can recommend lifestyle changes, regular monitoring, and timely treatment. Predictive models help in making informed decisions, optimizing resources, and improving patient outcomes, ultimately reducing the burden of diabetes on individuals and the healthcare system.

Data Exploration Insights

- **Key Observation**
  - Dataset Composition: The dataset consists of 6,000 rows and 7 columns.
  - The data includes patient_id, age, bmi, exercise_hours_per_week, smoker, blood_pressure, and the target variable diabetes.
  - The dataset contains 5,952 patients after removing outliers.
  - Class imbalance: 91% non-diabetic, 9% diabetic.
  - Age distribution: Diabetic patients are generally older; median age is higher than non-diabetic patients.
  - BMI analysis: Diabetic patients typically have higher BMI, often in the overweight or obese range.
  - Smoker status: 69.3% of patients are non-smokers.
  - Correlation analysis:
    - Strong positive correlation between age and blood pressure (0.71).
    - Moderate correlation between BMI and blood pressure (0.35).

- **Features Influencing Prediction**
  - BMI is the strongest predictor of diabetes risk.
  - Age is the next most influential factor.
  - Correlation analysis:
    - Diabetes positively correlates with BMI, age, and blood pressure.
    - Exercise hours and smoking status show minimal direct correlation with diabetes.
  - Logistic Regression feature importance:
    - BMI has the highest coefficient, contributing most to predictions.
    - Blood pressure has a moderate impact.
    - Exercise and smoking have negligible influence.

Fig 1 : Distribution of Diabetic and Non Diabetic in Dataset

```
counts = df['diabetes'].value_counts()
per    = df['diabetes'].value_counts(normalize=True) * 100
dist   = pd.DataFrame({'Count': counts, 'Percentage%': per.round(2)})
dist.index = ['No Diabetes (0)', 'Diabetes (1)']
print(dist)
```

```
                Count  Percentage%
No Diabetes (0)  5458        90.97
Diabetes (1)      542         9.03
```

Fig 2 : Smoker and Diabetes Distribution Visualization

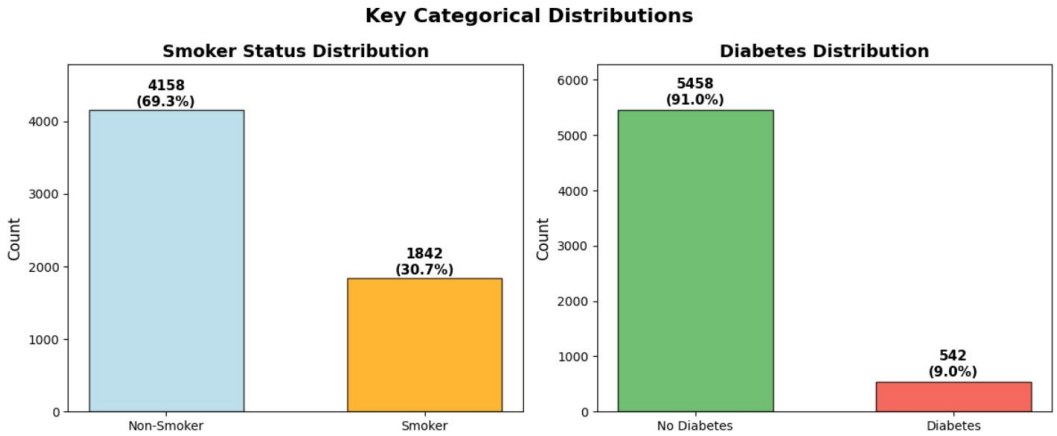
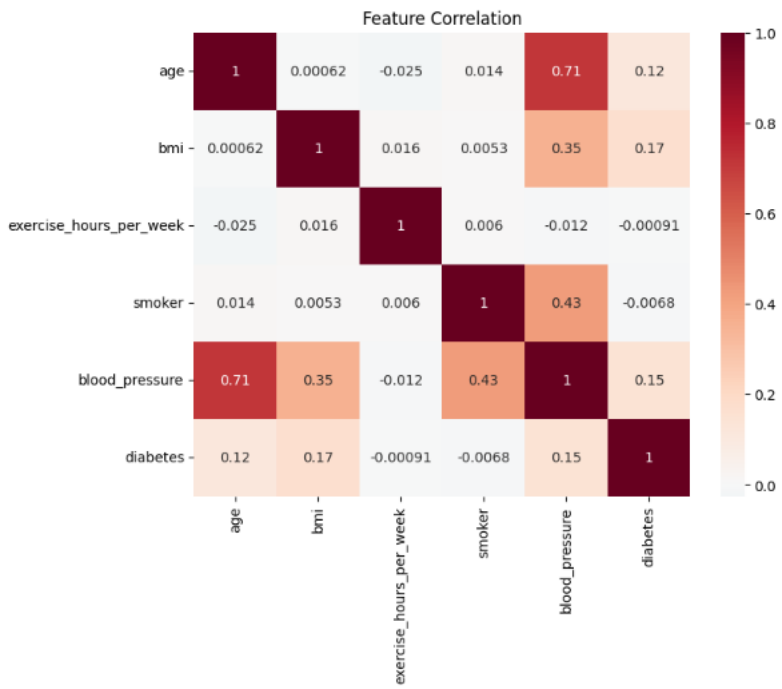


Fig 3 : Feature Co-Relation Heat Map



- ## Model Implementation
  - **Machine Learning Model - Logistic Regression**
    - **Features used (X):** age, BMI, exercise hours per week, smoker, blood pressure.
    - **Target (y)**: diabetes.
    - **Data split**: 80% training, 20% testing, with stratification to maintain class balance.
    - **Feature scaling**: StandardScaler applied for better model performance.
    - **Model training**: Logistic Regression with `class_weight='balanced'` to handle imbalanced data.
    - **Evaluation**: Accuracy, Precision, Recall, F1-score, and Confusion Matrix.
    - **Feature importance**: Positive coefficients increase diabetes risk, negative coefficients decrease it.

- **Neural Network Model**
  - **Features (X):** age, BMI, exercise hours per week, smoker, blood pressure
  - **Target (y):** diabetes
  - **Imbalance handling:** SMOTE applied to training data
  - **Data split:**
    - Train/Test = 80%/20% (stratified)
    - Train/Validation = 85%/15% (stratified)
  - **Feature scaling:** StandardScaler (fit on training set only)
  - **Model architecture:**
    - Input layer → Hidden layers: 64 → 32 → 16 (ReLU) → Output: 1 neuron (Sigmoid)
    - Regularization: L2 + Dropout (0.3, 0.2, 0.1)
  - **Training setup:** Adam optimizer (lr=0.0005), binary crossentropy loss, metrics: accuracy & AUC
  - **Training control:** EarlyStopping + ReduceLROnPlateau
  - **Prediction:** Probabilities thresholded at 0.5
  - **Evaluation:** Accuracy, Precision, Recall, F1-score, Confusion Matrix, Classification Report

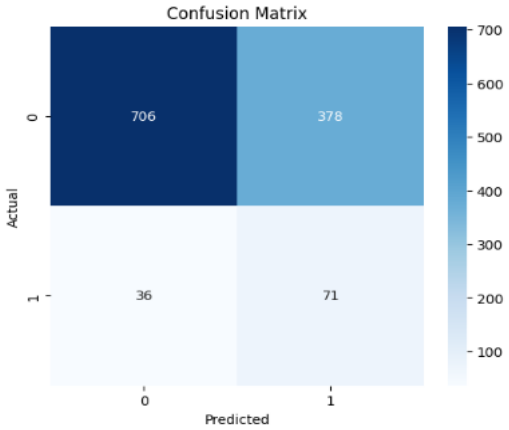
- ## Model Evaluation
  - Performance metrics for both models

Logistic Regression
- Accuracy : 65%
- Precision : 0.1581
- Recall : 0.6636
- F1 Score : 0.2554

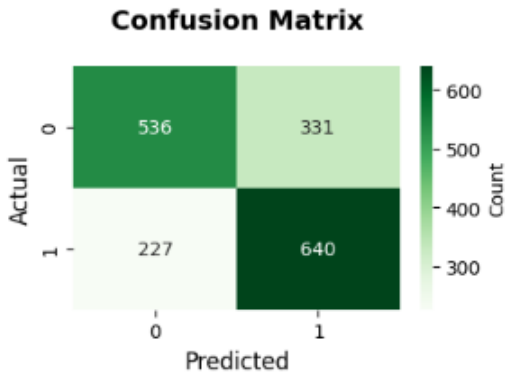
Neural Network Model
- Accuracy : 67 %
- Precision : 0.6591
- Recall : 0.7382
- F1 Score : 0.6964

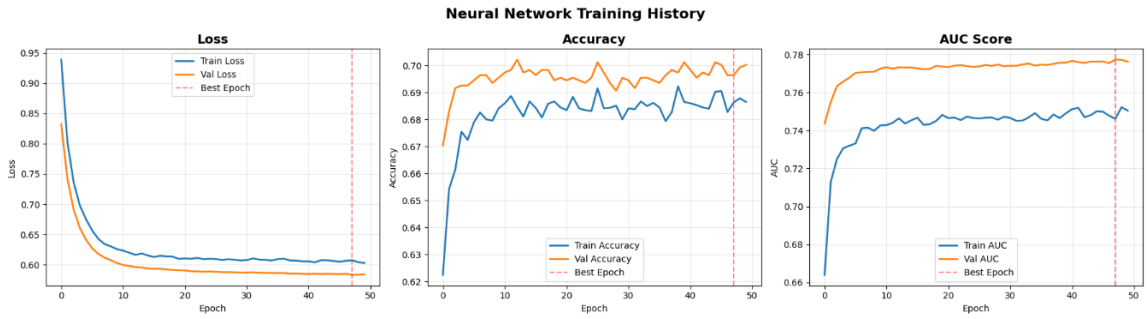
- Visualizations (graphs, confusion matrix)

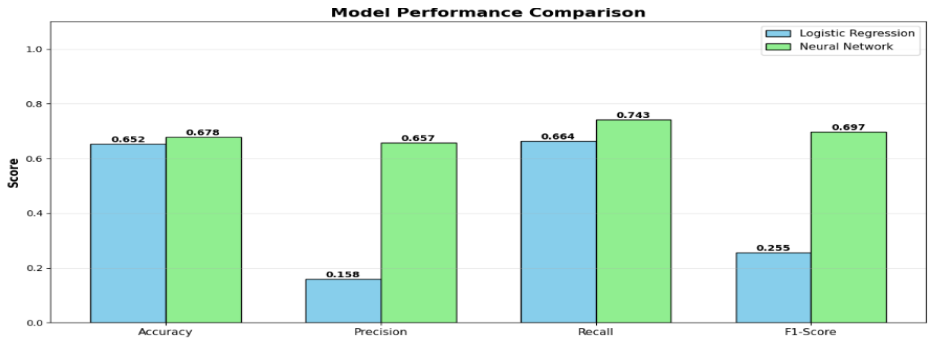
Logistic Regression -Confusion Matrix



Neural Network Model -Confusion Matrix







- ## Comparative Analysis
  - **Differences between the models**
    - **Logistic Regression (LR):** Linear model; predicts diabetes using a weighted sum of features + sigmoid. Easy to interpret using coefficients.
    - **Neural Network (NN):** Multi-layer non-linear model; learns complex relationships between features. Harder to interpret but more flexible.
    - **Training approach:** LR uses class_weight='balanced'; NN uses **SMOTE + class weights + regularization (Dropout, L2)** and validation-based training control (EarlyStopping, ReduceLROnPlateau).
  - **Reasons for accuracy (and metric) differences**
    - **Non-linearity:** NN can capture interactions (e.g., BMI + age combined effects) that LR may miss.
    - **Imbalanced data handling:** SMOTE and class weights in NN can improve **recall/F1** by learning minority-class patterns better.
    - **Overfitting control:** Dropout, L2, and early stopping help NN generalize; LR is simpler but may underfit if the boundary is not linear.
    - **Threshold effects:** Both models use a default **0.5** threshold; changing it can shift precision vs recall significantly.

- ## Recommendations for real-world use
  - **Choose LR** when interpretability and clinical explainability are required (clear feature impact, simpler deployment).
  - **Choose NN** when the priority is better overall detection/balance (often higher recall/F1), and you can support monitoring and explainability tools.
  - For screening use-cases, prioritize **recall** (reduce missed diabetes) and tune the decision threshold to manage false positives.